3.Vacuum Cleaner Problem

Aim

To simulate a Vacuum Cleaner Agent.

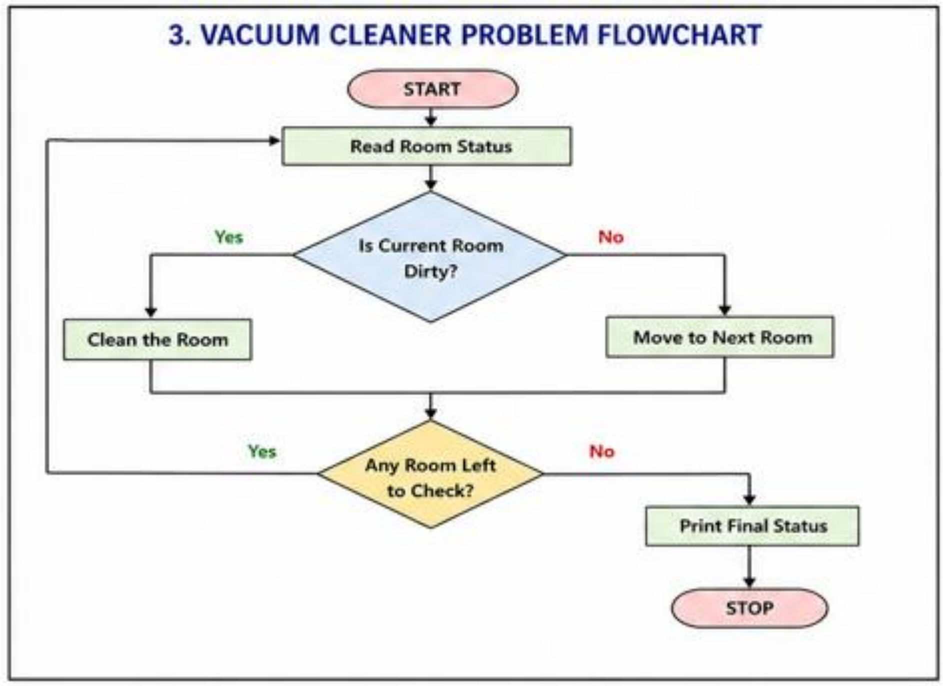
Objective

To clean all dirty rooms.

Algorithm

1. Check current location.
2. If dirty, clean.
3. Move to next room.
4. Repeat until all rooms are clean.

Flowchart

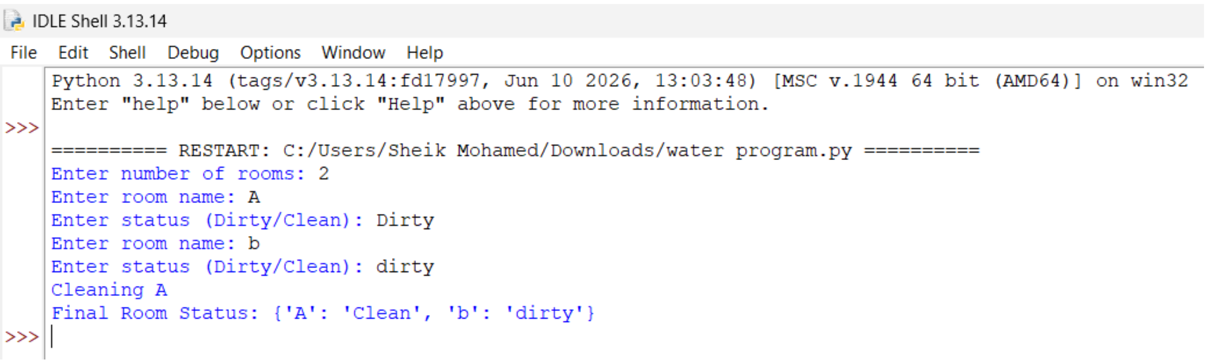


Python code:

```
rooms = {}


n = int(input("Enter number of rooms: "))


for i in range(n):
```

```
    room = input("Enter room name: ")
    status = input("Enter status (Dirty/Clean): ")
    rooms[room] = status

for room in rooms:
    if rooms[room] == "Dirty":
        print("Cleaning", room)
        rooms[room] = "Clean"

print("Final Room Status:", rooms)
```

output:



Result

All rooms were cleaned.

Conclusion

The Vacuum Cleaner Agent successfully achieved its goal.